

**Control for Nonlinear Effects of Baseline Confounders in Cross-Lagged Panel
Research: A Comparison of Latent Variable Approaches and Baseline
Covariate Adjusted SEM**

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Abstract

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1. Introduction

In psychology and related disciplines, a common research question is how two variables influence each other over time, that is, if one construct affects the other one time point later, and vice versa ([hamaker_within-between_2023](#)). Such causal relationships are also sometimes referred to as *reciprocal relationships*, *cross-lagged effects*, or *dynamics*. A popular approach to studying relationships of this type is with the use of panel data—where the same people are measured on the same variables at two or more measurement waves—which are subsequently analyzed with one of various cross-lagged panel models within the structural equation modeling (SEM) framework (Usami et al., [2019](#)). While parameter estimates of cross-lagged effects in these longitudinal SEM models are sometimes explicitly interpreted as causal effects (e.g., Orth et al., [2021](#); Zyphur et al., [2020](#)), their causal interpretation is more often implicit through suggestions for interventions and policy changes based on a study’s analysis results and the use of specific language (Grosz et al., [2020](#); Hamaker et al., [2020](#)). However, the validity of a causal interpretation of estimated effects relies on researchers identifying and correctly controlling for confounders of the causal effect of interest in their analysis; if important confounders remain unidentified (i.e., there exist unobserved confounders) or when the influence of observed confounders is not correctly modeled (i.e., the model is misspecified), estimates of causal effects might be biased, which can subsequently lead to erroneous conclusions and even harmful suggestions for interventions ([muthen_regression_2016](#); Murayama & Gfrörer, [2024](#)).

In practice, there are roughly two modeling approaches that are used for controlling for confounders that are stable over time—also referred to as *baseline* or *time-invariant* confounders. First, when baseline confounders are observed, they can be included in the

SEM model as predictors of the constructs of primary interest at each measurement occasion (Little, 2013). Doing so partials out the influence of the baseline covariates from the repeated outcome variables X and Y , such that estimated cross-lagged relationships between X and Y cannot be attributed to their shared covariates. Second, when baseline confounders remain unobserved, researchers can attempt to control for them through the inclusion of latent variables (**vink_manuscript_2025**). Many longitudinal SEM models, such as the dynamic panel model (DPM; Gische et al., 2021), the random intercept cross-lagged panel model (RI-CLPM; Hamaker et al., 2015), and the trait-state-error model (Kenny & Zautra, 2001) include such stable latent variables which, from a causal inference perspective, have the advantage of controlling for baseline confounders that have time-invariant effects on the outcomes X and Y over time even when these confounders are unobserved (Murayama & Gfrörer, 2024; Usami et al., 2019).

Both of these approaches, as well as a combined approach, have been shown to work well when the relationships between baseline confounders and the repeated outcomes can be assumed to be linear, and, in the case of the latent variable approach, when the effects of unobserved baseline confounders are invariant across time (**vink_manuscript_2025**). However, nonlinear relationships are common in psychology and related sciences—some examples involving typical baseline covariates include the relationship between age and criminal behavior in children (Keijsers et al., 2012), age and satisfaction with life (Darbonne et al., 2013), income and life satisfaction (Morris et al., 2021) and income and well-being (Kahneman & Deaton, 2010). When the nonlinear effects from baseline confounders are incorrectly modeled as linear there is a concern that both approaches break down (Bollen, 1989; Gische et al., 2021; Murayama & Gfrörer, 2024). Furthermore, in this situation, there is a lack of methodological guidance on how to appropriately account for baseline confounding. This gives rise to two research questions. First, how robust are common baseline confounding adjustment approaches for estimating cross-lagged effects in the presence of nonlinear relationships from baseline covariates?

Second, can we exploit nonlinear modeling techniques from the machine learning literature to correctly model and account for potentially nonlinear effects of baseline confounders in cross-lagged panel research? To investigate the first question, we perform a simulation study. For the second, we introduce a partially linear modeling approach to control for baseline confounders, which we call the baseline covariate adjusted (BCA) SEM approach. BCA SEM decouples statistical control for baseline covariates from the estimation of cross-lagged effects, thereby allowing for more flexibility in modeling the relationships between baseline covariates and constructs of interest with nonlinear parametric models and, in principle, nonparametric (machine learning) techniques.

This paper is structured as follows. In Section 2, we briefly explain the problem of baseline confounding using a substantive example. We then outline the common linear approaches to confounding adjustment in longitudinal SEM models. Section 3 then presents a simulation study to compare the performance of these approaches in the presence of nonlinear effects from baseline covariates. In Section 4 we introduce the BCA SEM method, describe its connection to the linear adjustment approaches, show how it can be combined with nonparametric (machine learning) techniques. A mathematical justification for the BCA SEM method is provided in Appendix B. In Section 5 we apply the linear approaches and the BCA SEM methods to the substantive example. Section 6 provides recommendations for empirical researchers, discusses limitations of the method, and suggests future research on BCA SEM. Annotated R code used for the simulations can be found in the online supplementary materials at

2. Current Approaches for Baseline Covariate Adjustment

Before describing current strategies to adjust for linear effects of baseline confounders and nonlinearity in the context of these models, we begin by introducing the problem of baseline confounding using a substantive example and a directed acyclic graph (DAG). This section also introduces notation that will be used throughout the paper.

2.1 Baseline Confounders in Cross-Lagged Panel Research

Suppose a group of researchers is interested in the reciprocal effects between perceived physical health and life satisfaction over time in a general population of Dutch adults. Specifically, the researchers want to know whether increased perceived physical health increases an individual's satisfaction with life one year later, and/or whether increased satisfaction with life increases later physical health. To investigate this, the researchers collect panel data, consisting of five waves of data on the two variables of interest, and a set of baseline covariates including demographic variables (e.g., age, gender, income), personality traits, and contextual variables such as the presence of the COVID-19 pandemic.

To aid in the decision of a correct analysis approach, the researchers represent the assumed underlying causal structure of the health-life satisfaction process in a DAG, as shown in Figure 1. Here, health at time point t is represented by X_t and life satisfaction at time point t is represented by Y_t . An arrow (also referred to as an *edge*) implies a direct causal effect of an exposure variable (also referred to as a *treatment* in the context of an experiment) to the outcome. The arrows from X_{t-1} to Y_t and from Y_{t-1} to X_t then represent the cross-lagged effects that are of primary interest from a causal perspective. The $\mathbf{C}$ represents a set of stable, baseline confounders that have causal effects on both repeatedly measured outcomes X_t and Y_t . To avoid cluttering, we have represented the causal effects of $\mathbf{C}$ by the dashed arrows.

The DAG in Figure 1 is a causal structure that is typically assumed, albeit implicitly so, to exist in much of the empirical research into cross-lagged effects. It already encodes specific structural (also sometimes referred to as *causal* or *conditional independence*) assumptions, including (a) an absence of contemporaneous, lag-0 effects such as $X_t \rightarrow Y_t$ or $Y_t \rightarrow X_t$; (b) an absence of delayed, lag-2 or later effects such as $X_{t-2} \rightarrow Y_t$ or $Y_{t-2} \rightarrow X_t$; (c) an absence of time-varying confounders; and (d) that X_t and Y_t are constant between two measurement occasions, that is, that the values that X and Y take

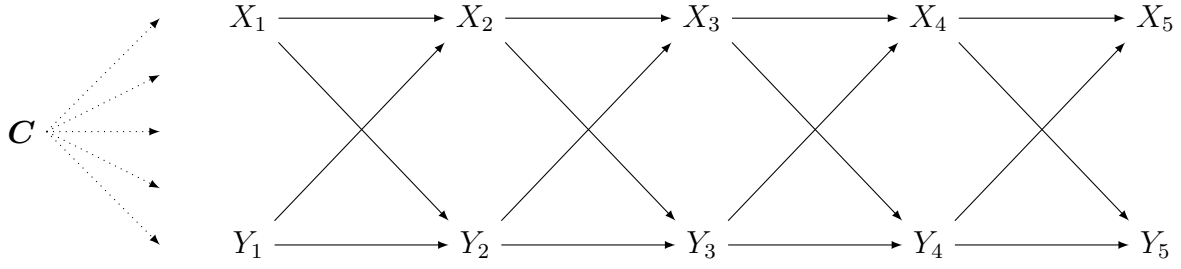


Figure 1

The Effects of Health and Life Satisfaction on Each Other Over Time. $\mathbf{C}$ Represents Multiple Baseline Confounders That Affect Health and Life Satisfaction at All Future Timepoints. Subscripts Indicate the Timepoints.

on at time point t remain unchanged until just before time point $t + 1$. For the purpose of this paper, we will assume that these structural assumptions hold in the coming sections. In the discussion in Section 7, we briefly discuss how the discussions in this paper change when some of the assumptions are weakened/deemed unrealistic. However, one aspect that the DAG does not make any assumptions about, and which is the focus of this paper, is the functional form of its causal dependencies. For example, the causal effects of $\mathbf{C}$ in the DAG in Figure 1 can follow any functional form, potentially including higher-order terms (e.g., quadratic or cubic effects), interactions, sinusoidal patterns, or even discontinuous jumps. This paper studies how well cross-lagged effects can be estimated with commonly applied fully linear longitudinal SEM models when the cross-lagged effects are correctly assumed to be linear, but the causal effects of the baseline confounders $\mathbf{C}$ are themselves nonlinear (i.e., when the statistical model is thus partially misspecified).

2.2 Adjustment for Baseline Confounding in Cross-Lagged Panel Modeling

In the longitudinal SEM literature there are two general and common approaches to adjust for baseline confounding. Both of these approaches have been described for cases where linearity for the relationships between the baseline confounders and the outcomes can be assumed, but they differ in their assumptions about which baseline confounders need to be observed (i.e., they rely on different versions of the “no unmeasured confounding” assumption). The first approach involves adding observed baseline covariates

as predictors for both constructs of interest in a longitudinal SEM model (Bollen, 1989; Little, 2013). In the context of the substantive example, this implies that researchers should measure variables such as age, gender, social economic status, and personality, and regress health X_t and anxiety Y_t at each time point on the baseline covariates, as well as on lagged versions of themselves, X_{t-1} and Y_{t-1} . A cross-lagged panel model (CLPM) is an example of a popular and fully linear longitudinal SEM model (Rogosa, 1980). When extended to include baseline covariates, it is given by

$$X_t = \gamma_{XX,t} X_{t-1} + \gamma_{XY,t} Y_{t-1} + \gamma_{XC,t}' \mathbf{C} + \epsilon_{X,t}, \quad (1)$$

$$Y_t = \gamma_{YY,t} Y_{t-1} + \gamma_{YX,t} X_{t-1} + \gamma_{YC,t}' \mathbf{C} + \epsilon_{Y,t}, \quad (2)$$

where the grand means are assumed to be 0, $\gamma_{XC,t}$ and $\gamma_{YC,t}$ are the effects of the baseline covariates on X_t and Y_t , respectively, and $\gamma_{XY,t}$ and $\gamma_{YX,t}$ are the the cross-lagged effects from health to anxiety and from anxiety to health. For this strategy to result in unbiased estimates of the cross-lagged effects, we have rely on the structural assumptions as introduced in the previous section, and additionally have to assume that all baseline confounders are measured and included in $\mathbf{C}$ and that their relationship to the observed outcomes is indeed linear (Bollen, 1989).

The second approach concerns extensions of the traditional CLPM with latent variables, resulting in currently popular longitudinal SEM models such as the random intercept cross-lagged panel model (RI-CLPM; Hamaker et al., 2015) and the dynamic panel model (DPM; Gische et al., 2021). The RI-CLPM is represented by

$$X_t = I_X + W_{X,t}, \quad (3)$$

$$Y_t = I_Y + W_{Y,t}, \quad (4)$$

$$W_{X,t} = \gamma_{XX,t} W_{X,t-1} + \gamma_{XY,t} W_{Y,t-1} + \epsilon_{X,t} \quad (5)$$

$$W_{Y,t} = \gamma_{YY,t} W_{Y,t-1} + \gamma_{YX,t} W_{X,t-1} + \epsilon_{Y,t}, \quad (6)$$

and is based on decomposing the observed scores of an individual into a stable person-specific deviation from the grand mean (i.e., a random intercept, I_X and I_Y), and a temporal deviation from this stable component ($W_{X,t}$ and $W_{Y,t}$). These temporal deviations make up the *within-person* part of the RI-CLPM; the dynamics are modeled using autoregressions and cross-lagged regressions for these within-person components with innovations ϵ representing the unexplained part. This model results in unbiased estimates of the cross-lagged effects when both the effects of the baseline confounders on the outcomes and the lagged effects among the within-components are time-invariant (Usami et al., 2019). Analytical work by Murayama and Gfrörer (2024) and Usami et al. (2019) and simulations by **vink_manuscript_2025<empty citation>** show that under these assumptions, the RI-CLPM correctly controls for a large number of baseline confounders with linear effects on the outcomes, even when these confounders are unobserved.

While the RI-CLPM accounts for stable factors, it deviates from the DAG in Figure 1 in that the dynamics are not included between the observed variables, but between the within-person components. The DPM, however, is much more closely related to the DAG in Figure 1, and is given by

$$X_t = \eta_X + \gamma_{XX,t} X_{t-1} + \gamma_{XY,t} Y_{t-1} + \epsilon_{X,t}, \quad (7)$$

$$Y_t = \eta_Y + \gamma_{YX,t} X_{t-1} + \gamma_{YY,t} Y_{t-1} + \epsilon_{Y,t}. \quad (8)$$

The model includes a latent variable for each of the two repeatedly measured variables separately (η_X and η_Y), as does the RI-CLPM, but the autoregressive and cross-lagged relations are modeled between the observed variables themselves rather than between the within-person deviations. Furthermore, η_X and η_Y are often called *accumulating factors* or *additive random intercepts* since their influence on X and Y builds up over time due to their indirect effects through previous timepoints. For unbiased estimates of the cross-lagged effects using the DPM, one has to assume that the unobserved confounders are

only baseline confounders that have time-invariant relationships with the outcomes X and Y . In contrast to the RI-CLPM, the DPM does not require the assumption of time-invariant lagged effects to control for unobserved baseline confounders.

While the fact that not all baseline confounders need to be measured is an appealing feature of the DPM and the RI-CLPM, the disadvantage is that we have to make assumptions about the structure of the unmeasured confounders, specifically that the effects of these confounders on the outcomes X and Y are time-invariant. This assumption does not have to be made when baseline confounders are measured and included as predictors in the SEM model as in the first approach. However, the two adjustment approaches are not mutually exclusive. Baseline covariates can be included as predictors of the observed constructs X_t and Y_t in the DPM and the RI-CLPM to control for observed confounders, while relying on the latent factors in these models to control for unobserved baseline covariates under the assumption that they have time-invariant effects (vink__manuscript__2025). Even so, all these models consist essentially of only linear regressions. When effects from baseline confounders on the outcomes are nonlinear, there is a concern that estimates of cross-lagged effects will be biased (Bollen, 1989; Gische et al., 2021; Murayama & Gfrörer, 2024).

2.3 Nonlinearity in SEM Models

To systematically investigate the impact of nonlinear effects of baseline confounders, we first describe different forms of nonlinearity that are considered in psychology and related sciences, and how they play a role in SEM. One form of nonlinear effects that is commonly considered in empirical research, especially when it concerns relationships with age, are quadratic or cubic effects (e.g., Darbonne et al., 2013; Keijsers et al., 2012). Such nonlinear effects are often modeled in SEM models by including quadratic or cubic terms of specific predictors as additional covariates. Another popular form of nonlinearity concerns interactions. Especially when interest is in moderation, interaction terms between two predictors are included as additional covariates in the model, similarly to how quadratic or

cubic effects are modeled. Furthermore, sometimes nonlinearities involve seasonal effects ([haqiqatkhah_dynamics_preprint](#)) or sudden jumps such as These forms of nonlinearity are not exclusive and can be combined to complex relationships between variables. In the context of our empirical example, we can denote the relationship between the repeatedly measured outcomes X_t and Y_t and the baseline confounders $\mathbf{C}$ generally as

$$\mathbb{E}[X_t|\mathbf{C}] = f_{X,t}(\boldsymbol{\theta}_{X,t}, \mathbf{C}), \quad (9)$$

and

$$\mathbb{E}[Y_t|\mathbf{C}] = f_{Y,t}(\boldsymbol{\theta}_{Y,t}, \mathbf{C}), \quad (10)$$

where $f_{X,t}(\cdot)$ and $f_{Y,t}(\cdot)$ are functions that describe how X_t and Y_t are influenced by some, potentially nonlinear combination of observed baseline covariates and the parameters in $\boldsymbol{\theta}_{X,t}$ and $\boldsymbol{\theta}_{Y,t}$, respectively.

To estimate cross-lagged effects in the presence of potentially nonlinear effects from baseline covariates, we consider two approaches: A fully parametric approach and a semi-parametric approach. In the fully parametric approach, researchers themselves determine apriori what the correct functional form is for the relationships between $\mathbf{C}$ and the outcomes (i.e., researchers determine what the functions $f_{X,t}(\cdot)$ and $f_{Y,t}(\cdot)$ should be). Such decisions then have to be based on theory and the any resulting estimates from the model rely on this assumed functional form being correct. As long as the functional form can be described by a polynomial—that is, a linear sum of potentially nonlinear terms such as the quadratic term X^2 or the two-way interaction X_1X_3 —the nonlinear terms that make up the polynomial can be computed from the data and added as additional covariates to the model. When the assumed functional form is wrong, the model is said to be misspecified, which not only leads to bias in the effects of the baseline confounders $\boldsymbol{\theta}_{X,t}$ and $\boldsymbol{\theta}_{Y,t}$, but also in parameters in other parts of the model such as the cross-lagged effects of primary interest. A major problem with this fully parametric approach is that you will not

be able to detect misfit due to omitted nonlinear terms in the fit indices of SEM model. Decisions regarding the functional form of the $\mathbf{C}$ and outcome relationships thus have to be fully determined by the researcher when using this approach.

There are, however, a couple of cases where misspecification of the functional form does not translate in bias in the SEM model. These include when quadratic terms of variables that are centered at zero (**irmer__**), two-way interactions (**muthe__regression__2016**). The reason is that SEM is based on analyzing the variance-covariance matrix of the observed data. These specific instances of nonlinear terms do not change the covariances between the other variables. Hence, in the context of our empirical example, when a quadratic relationship between $\mathbf{C}$ and the outcomes is incorrectly specified to be linear, this does not translate to bias in other parts of the model in the cross-lagged effects between X and Y .

In the context of this fully parametric approach, several research questions arise. First, how badly does a functionally misspecified baseline confounder-outcome relationship affect the relationships between outcomes themselves. Furthermore, can the inclusion of a latent variable such as in the RI-CLPM or the DPM alleviate the impact of misspecified nonlinear relationships? These research questions are studied in the simulation study in Section 3.

For situations where researchers are unsure about what the correct functional form $f_{X,t}(\cdot)$ and $f_{Y,t}(\cdot)$ should be, nonparametric (machine learning) techniques might be employed to attempt to infer the correct functional form from the data. To exploit their power, we first need to find a way to connect cross-lagged panel modeling using SEM with machine learning techniques. For this, we describe a new, two-step approach that we refer to as baseline covariate adjusted (BCA) SEM. This approach can be regarded as a *partially linear* approach as it uses machine learning techniques to control for baseline confounder in a nonparametric way, but linear SEM models to subsequently estimate the cross-lagged effects. We will introduce this method in Section 4.

BCA-nonlinear

$X_0 + X_1 + X_2 + X_3 + X_4 + X_5 \sim C + C^2$

$Y_0 + Y_1 + Y_2 + Y_3 + Y_4 + Y_5 \sim C + C^2$

Identification issue: If $C_2 = C^2$ and C has a stable (or no) effect on X and Y over time, then C and the latent variables are not distinguishable, leading to a lack of identification of effects. Instead of estimating the effect of C onto X and Y as zero, the model will produce distorted effects, and the latent variances of F_X and F_Y will be incorrect. If I fix the effect of C to zero, then identification is gained and everything works fine.

Interestingly, this issue does not occur if I compute true residuals, highlighting the flexibility of that approach. This might also affect our other project, unless we are already computing “true” residuals there. It might be special in that case since C is uncorrelated with C^2 (if C is normally distributed), so the effect of C could otherwise mix up with the latent variables.

Fixed-effect version:

BCA-nonlinear

$X_0 + X_1 + X_2 + X_3 + X_4 + X_5 \sim 0 \cdot C + C^2$

$Y_0 + Y_1 + Y_2 + Y_3 + Y_4 + Y_5 \sim 0 \cdot C + C^2$

With this restriction, I obtain the same results as with “true” residualization. However, the model fit seems poor under this restriction, suggesting an identification issue. We should discuss this and check for implications for the other project.

4. The BCA SEM Approach

If researchers are not comfortable with making assumptions about the functional form of the influences of the baseline confounders themselves, a semi-parametric or nonparametric approach can be taken instead. In such an approach, the functional form of

particular relationships do not have to be assumed a priori by research, but can instead be derived from the data. This is one of advantages of machine learning approaches, and one of the motivations for incorporating machine learning into methodology for causal inference. In this context, the question is how such nonparametric machine learning techniques can be combined with SEM models including latent variables, for the purpose of controlling for nonlinear effects of baseline confounders. For this, we introduce a new approach in the next Section.

In this section, we introduce the new BCA SEM approach. Compared to the fully-linear longitudinal SEM models that were introduced in Section 2.2, BCA SEM does not necessarily assume that the relationships between the baseline confounders and the outcomes X_t and Y_t are linear, and therefore is, in principle, more robust to model misspecification and bias in the cross-lagged effects. BCA SEM does still assume linearity for the lagged relationships between the outcomes, and can therefore be regarded as a partially-linear or semi-parametric approach to cross-lagged panel modeling.

We will apply the new method to settings where we have available a set of observed baseline covariates that we want to control for, and additionally want to control for remaining unobserved baseline covariates with a latent variable, either an accumulating factor or a random intercept factor. The full set of baseline confounders (both observed and unobserved) is denoted by $\mathbf{C}$ as in the DAG of Figure 1, while the *observed* subset of baseline covariates is denoted by $\mathbf{C}_O$ and the *unobserved* subset of baseline covariates by $\mathbf{C}_U$. In Section 3.1, we apply the BCA SEM procedure to the DPM and the RI-CLPM, and go through the procedure of fitting a BCA DPM and a BCA RI-CLPM to panel data. A mathematical justification for the BCA SEM approach can be found in Appendix B. Finally, Annotated R code for the fitting procedure of a BCA DPM and a BCA RI-CLPM can be found in the online supplementary materials at

3.1 Fitting the BCA DPM and BCA RI-CLPM

The fitting procedure for both the BCA DPM and the BCA RI-CPLM consists of two steps. The first step is identical for both models and involves creating BCA scores for X_t and Y_t by removing the shared variance of these observed variables with the observed time-invariant covariates $\mathbf{C}_O$. The newly created BCA scores are actually the residuals obtained from a regression of X_t and Y_t on $\mathbf{C}_O$, and are denoted by $\mathcal{X}_t$ and $\mathcal{Y}_t$, respectively. Very generally, the regression of X_t on observed covariates can be expressed as

$$\mathbb{E}[X_t|\mathbf{C}_O] = f_{X,t}(\boldsymbol{\theta}_{X,t}, \mathbf{C}_O), \quad (11)$$

and the regression of Y_t on observed covariates as

$$\mathbb{E}[Y_t|\mathbf{C}_O] = f_{Y,t}(\boldsymbol{\theta}_{Y,t}, \mathbf{C}_O). \quad (12)$$

In practice, researchers commonly assume a particular functional form for these functions apriori. For example, a frequent decision is to assume linearity, such that

$$\mathbb{E}[X_t|\mathbf{C}_O] = f_{X,t}(\boldsymbol{\theta}_{X,t}, \mathbf{C}_O) = \boldsymbol{\beta}_{X,t}'\mathbf{C}_O, \quad (13)$$

$$\mathbb{E}[Y_t|\mathbf{C}_O] = f_{Y,t}(\boldsymbol{\theta}_{Y,t}, \mathbf{C}_O) = \boldsymbol{\beta}_{Y,t}'\mathbf{C}_O, \quad (14)$$

where $\boldsymbol{\beta}_{X,t}$ is a vector containing linear regression coefficients for the relation between all observed covariates in $\mathbf{C}_O$ and X_t , and $\boldsymbol{\beta}_{Y,t}$ is a vector containing linear regression coefficients for the relation between all observed covariates in $\mathbf{C}_O$ and Y_t . The apostrophe $'$ denotes the transpose of these vectors. Furthermore, the intercepts of the regression models are included in $\boldsymbol{\beta}_{X,t}$ and $\boldsymbol{\beta}_{Y,t}$. These linear regression models need to be fitted for X_t and Y_t at each time point separately; this can be done using ordinary least squares. The BCA scores are then computed as the difference between the observed values of X_t and Y_t and their expected values based on the observed time-invariant covariates included in $\mathbf{C}_O$,

that is,

$$\mathcal{X}_t = X_t - \mathbb{E}[X_t | \mathbf{C}_O], \quad (15)$$

$$\mathcal{Y}_t = Y_t - \mathbb{E}[Y_t | \mathbf{C}_O]. \quad (16)$$

Note that since $\mathbf{C}_O$ only includes baseline covariates, the BCA scores are *not* adjusted for X and Y at previous time points, nor for the unobserved time-invariant confounders in $\mathbf{C}_U$.

The linearity assumption in the models in Equations 13 and 14 can be evaluated based on theory, prior research, or empirically tested with plots of the linearly created BCA scores against the baseline covariates, so-called residual plots. When the linearity assumption concerning the baseline covariate- X_t and baseline covariate- Y_t relationships is deemed unrealistic, the functions $f_{X,t}(\cdot)$ and $f_{Y,t}(\cdot)$ can be extended to include nonlinear terms. As explained in Section 2.3, one option is to take a nonlinear parametric approach in which researchers make apriori assumptions about the correct functional form. For example, quadratic or cubic terms can be added to the regression models in Equations 11 and 12 such that

$$\mathbb{E}[X_t | \mathbf{C}_O] = f_{X,t}(\boldsymbol{\theta}_{X,t}, \mathbf{C}_O) = \boldsymbol{\beta}'_{X,t} \mathcal{P}_{d,k}(\mathbf{C}_O) \quad (17)$$

$$\mathbb{E}[Y_t | \mathbf{C}_O] = f_{Y,t}(\boldsymbol{\theta}_{Y,t}, \mathbf{C}_O) = \boldsymbol{\beta}'_{Y,t} \mathcal{P}_{d,k}(\mathbf{C}_O), \quad (18)$$

where $\mathcal{P}_{d,k}$ generates all polynomials up until order d and including k -way interactions. For quadratic terms where $d = 2$, $k = 2$ -way interactions, and with only two baseline confounders observed such that $\mathbf{C}_O$ consists of $\mathbf{C}_O = (C_1, C_2)'$, the term $\mathcal{P}_{2,2}(\mathbf{C}_O)$ is $\mathcal{P}_{2,2}(\mathbf{C}_O) = (1, C_1, C_2, C_1^2, C_2^2, C_1 C_2, C_1^2 C_2, C_1 C_2^2, C_1^2 C_2^2)'$ such that $\boldsymbol{\beta}'_{X,t} \mathcal{P}_{2,2}(\mathbf{C}_O) = \beta_{0,X,t} + \beta_{1,X,t} C_1 + \beta_{2,X,t} C_2 + \dots + \beta_{10,X,t} C_1^2 C_2^2$. Alternatively, researchers can forgo making any apriori assumptions about the functional form, and take a more data-driven approach. That is, a nonparametric approach can be taken by employing machine learning methods to determine the functional forms $f_{X,t}(\cdot)$ and $f_{Y,t}(\cdot)$ based on

data. This option, in principle, expands the BCA SEM approach beyond the nonlinear polynomial relationships in Equations 17 and 18 to the even more general relationships. However, naive applications of machine learning methods can result in overfitting of models and its biases.

After the creation of the BCA scores in the first step, we fit a longitudinal SEM model to these BCA scores in the second step. The BCA DPM is then given by

$$\mathcal{X}_t = \eta_{\mathcal{X}} + \gamma_{\mathcal{X}\mathcal{X},t} \mathcal{X}_{t-1} + \gamma_{\mathcal{X}\mathcal{Y},t} \mathcal{Y}_{t-1} + \zeta_{\mathcal{X},t}, \quad (19)$$

$$\mathcal{Y}_t = \eta_{\mathcal{Y}} + \gamma_{\mathcal{Y}\mathcal{X},t} \mathcal{X}_{t-1} + \gamma_{\mathcal{Y}\mathcal{Y},t} \mathcal{Y}_{t-1} + \zeta_{\mathcal{Y},t}. \quad (20)$$

In parallel to the fully linear DPM, the BCA scores $\mathcal{X}_1$ and $\mathcal{Y}_1$ at the first time point are not modeled and thus treated as exogenous variables. $\eta_{\mathcal{X}}$ and $\eta_{\mathcal{Y}}$ are latent variables that have $\mathcal{X}_t$ and $\mathcal{Y}_t$ at each time point as indicators, respectively. These variables have factor loadings of 1 and are the accumulating factors in a DPM (Usami et al., 2019). $\gamma_{\mathcal{X}\mathcal{X},t}$ and $\gamma_{\mathcal{Y}\mathcal{Y},t}$ are the autoregressive effects of $\mathcal{X}_t$ and $\mathcal{Y}_t$. $\gamma_{\mathcal{Y}\mathcal{X},t}$ is the cross-lagged effect of $\mathcal{X}_{t-1}$ to $\mathcal{Y}_t$ and $\gamma_{\mathcal{X}\mathcal{Y},t}$ is the cross-lagged effect of $\mathcal{Y}_{t-1}$ to $\mathcal{X}_t$. $\zeta_{\mathcal{X},t}$ and $\zeta_{\mathcal{Y},t}$ are the innovations for $\mathcal{X}_t$ and $\mathcal{Y}_t$, respectively. In addition to the structural relations in Equations 19 and 20, the BCA DPM includes covariances between the initial BCA scores and the accumulating factors $\eta_{\mathcal{X}}$ and $\eta_{\mathcal{Y}}$.

In estimating the causal effect of X on Y at the next time point (and vice versa), the above procedure statistically controls for *observed* time-invariant confounders in Step 1 through creation of the BCA scores, and additionally controls for *unobserved* time-invariant confounders through the inclusion of the accumulating factors $\eta_{\mathcal{X}}$ and $\eta_{\mathcal{Y}}$. The resulting estimators $\hat{\gamma}_{\mathcal{X}\mathcal{Y}}$ and $\hat{\gamma}_{\mathcal{Y}\mathcal{X}}$ are unbiased for the cross-lagged effects between X and Y when the regression models in Equations 11 and 12 of Step 1 are correct and when the *unobserved* time-invariant confounders ($\mathbf{C}_U$) only have effects on X_t and Y_t that are stable over time. When the effects of unobserved time-invariant confounders are nonlinear

or varying over time, then the inclusion of an accumulating factor is not sufficient and bias in estimates of the cross-lagged effects is expected to occur (Murayama & Gfrörer, 2024).

To obtain standard errors for the parameters of the BCA DPM, one cannot simply interpret the directly available maximum likelihood-based standard errors from SEM-software in Step 2. These standard errors are conservative (i.e., too small) because they ignore the uncertainty in the creation of the BCA scores in Step 1 of the procedure. Consequently, confidence intervals based on these standard errors will be too narrow, p -values will be too small, and the Type I error rate will thus be inflated. To obtain correct standard errors, a bootstrap procedure of the entire fitting procedure can be used. This implies that multiple bootstrap samples need to be drawn, and for each bootstrap sample the two-step procedure as outlined above needs to be run. The online supplementary materials contain annotated R-code to help researchers with implementing this procedure.

Alternatively, a BCA RI-CLPM can be fitted to the data. In Step 2, the BCA scores would then be decomposed as

$$\mathcal{X}_t = I_{\mathcal{X}} + W_{\mathcal{X},t}, \quad (21)$$

$$\mathcal{Y}_t = I_{\mathcal{Y}} + W_{\mathcal{Y},t}. \quad (22)$$

$I_{\mathcal{X}}$ and $I_{\mathcal{Y}}$ are latent variables referred to as random intercepts, capturing stable, between-person differences in the residuals over time. The key difference between these latent variables and $\eta_{\mathcal{X}}$ and $\eta_{\mathcal{Y}}$ from the BCA DPM is that their impact is restricted to a single time point. Hence, $I_{\mathcal{X}}$ and $I_{\mathcal{Y}}$ do not have an accumulating effect on the BCA scores $\mathcal{X}_t$ and $\mathcal{Y}_t$ as do $\eta_{\mathcal{X}}$ and $\eta_{\mathcal{Y}}$, which changes their interpretation (usami2019). $W_{\mathcal{X},t}$ and $W_{\mathcal{Y},t}$ are individual- and time-specific latent variables, commonly referred to as within-person components, capturing the deviations of individuals from their expected BCA scores based on the random intercepts.

The structural relationships between the variables are then modeled between the

within-components as

$$W_{\mathcal{X},t} = \delta_{\mathcal{X}\mathcal{X},t}W_{\mathcal{X},t-1} + \delta_{\mathcal{X}\mathcal{Y},t}W_{\mathcal{Y},t-1} + \xi_{\mathcal{X},t}, \quad (23)$$

$$W_{\mathcal{Y},t} = \delta_{\mathcal{Y}\mathcal{X},t}W_{\mathcal{X},t-1} + \delta_{\mathcal{Y}\mathcal{Y},t}W_{\mathcal{Y},t-1} + \xi_{\mathcal{Y},t}. \quad (24)$$

Here, $\delta_{\mathcal{X}\mathcal{X},t}$ and $\delta_{\mathcal{Y}\mathcal{Y},t}$ are the autoregressive effects that capture moment-to-moment stability in the within-components. $\delta_{\mathcal{X}\mathcal{Y},t}$ and $\delta_{\mathcal{Y}\mathcal{X},t}$ represent the cross-lagged effects from $W_{\mathcal{Y},t-1}$ to $W_{\mathcal{X},t}$ and from $W_{\mathcal{X},t-1}$ to $W_{\mathcal{Y},t}$ at the next occasion, respectively. $\xi_{\mathcal{X},t}$ and $\xi_{\mathcal{Y},t}$ are the innovations for the within-components. When the DAG in Figure 1 is true, the estimators $\hat{\delta}_{\mathcal{X}\mathcal{Y},t}$ and $\hat{\delta}_{\mathcal{Y}\mathcal{X},t}$ are unbiased for the cross-lagged effects of X on Y (and vice versa) when: (a) the regression models in Equations 11 and 12 are correct; (b) the unobserved time-invariant confounders $\mathbf{C}_U$ only have stable, linear effects on X_t and Y_t over time; and (c) the lagged effects $\delta_{\mathcal{X}\mathcal{X},t}$, $\delta_{\mathcal{Y}\mathcal{Y},t}$, $\delta_{\mathcal{X}\mathcal{Y},t}$, and $\delta_{\mathcal{Y}\mathcal{X},t}$ are stable over time (**usami2019**). In this approach, the confounding effect of observed time-invariant covariates is controlled for in Step 1 with the creation of the BCA scores and the confounding effect of unobserved time-invariant covariates with stable effects over time is controlled for by the random intercept factor. Similarly to the BCA DPM, bootstrapping is recommended to obtain valid standard errors for the parameters of the BCA RI-CLPM.

Appendix B contains the mathematical justification for the BCA SEM procedure as outlined above. The proof shows that researchers can confidently apply BCA SEM to control for baseline covariates with nonlinear effects on the constructs of interest, as long as the conditional expectations in Equations 11 and 12 are correctly specified. Those conditional expectations may be estimated with nonlinear parametric models, non-parametric regressions, or flexible machine-learning algorithms, provided the chosen method correctly captures the relevant dependency structure. Such flexibility is critical in practice, because real-world data often depart from linearity through interactions, threshold effects, or other complex patterns. When, nevertheless, linearity is assumed for this

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